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Merrivoss Bioanalysis Inc

Matrix acceptance sign-off

Bioanalytical QA, signed copy

Merrivoss Bioanalysis Inc, Madison

Prepared by: Helen Marchwood, Bioanalytical Quality Manager

Date: 09/13/2011

1 Purpose

This sign-off records the matrix acceptance rules that the bioanalysis group applies to tandem-MS runs from this date. It replaces the working note circulated in the spring and should be filed with the current assay worksheet template.

2 Calibration checks

Each run carries a full calibration curve prepared in the matrix of the study samples. Curves are fitted with a weighted linear regression. A calibration point is only used if it back-calculates within the accepted window on the day of the run.

Table 1 Calibration and matrix checks applied to each run

Check	Applied to	Action if it fails
Curve fit and back-calculation	Every calibration point	Point excluded and curve refitted; the exclusion is written on the worksheet
Blank matrix response	Each batch, before the first standard	Batch held until the carry-over source is identified
Internal standard response	Every sample	Sample reanalysed; not corrected by arithmetic
Quality control placement	Start, middle and end of each batch	Batch rejected and repeated if the run-level QC fails

Transition multiples are confirmed against the reference standard before each new analyte card is opened. Where two transitions disagree beyond the tolerance on the card, the quantitation transition is used and the discrepancy is noted in the analyst's comments.

3 Batch review responsibilities

The analyst prepares the batch and completes the first review. A second scientist, not involved in the run, checks the calibration, QC and internal standard data before the batch is released to the study director.

- Analyst: batch preparation, first review, worksheet entries
- Second reviewer: independent check of calibration and QC, signed on the worksheet
- Mass-spectrometry manager: instrument status and any tuning changes
- Study director: release of data to the study report, after matrix acceptance is recorded

Reviews are signed and dated on the worksheet in ink. Electronic reviews are accepted only where the LIMS audit trail is available for the same run.

4 Control lot reference

Quality control samples are prepared from the control lot held in the matrix freezer. The lot reference on the worksheet must match the freezer log before the batch is started. A new control lot is only put into use after it has been checked against the outgoing lot in at least one run.

Correction to the control lot reference in the draft of this sign-off has been made by hand on the signed copy and is initialled below.

5 Sign-off

Role	Name	Date
Bioanalytical Quality Manager	Helen Marchwood	09/13/2011
DMPK Lead	Matteo Rinalvesi	09/13/2011
Mass-Spectrometry Manager	Javier Quintana	09/14/2011